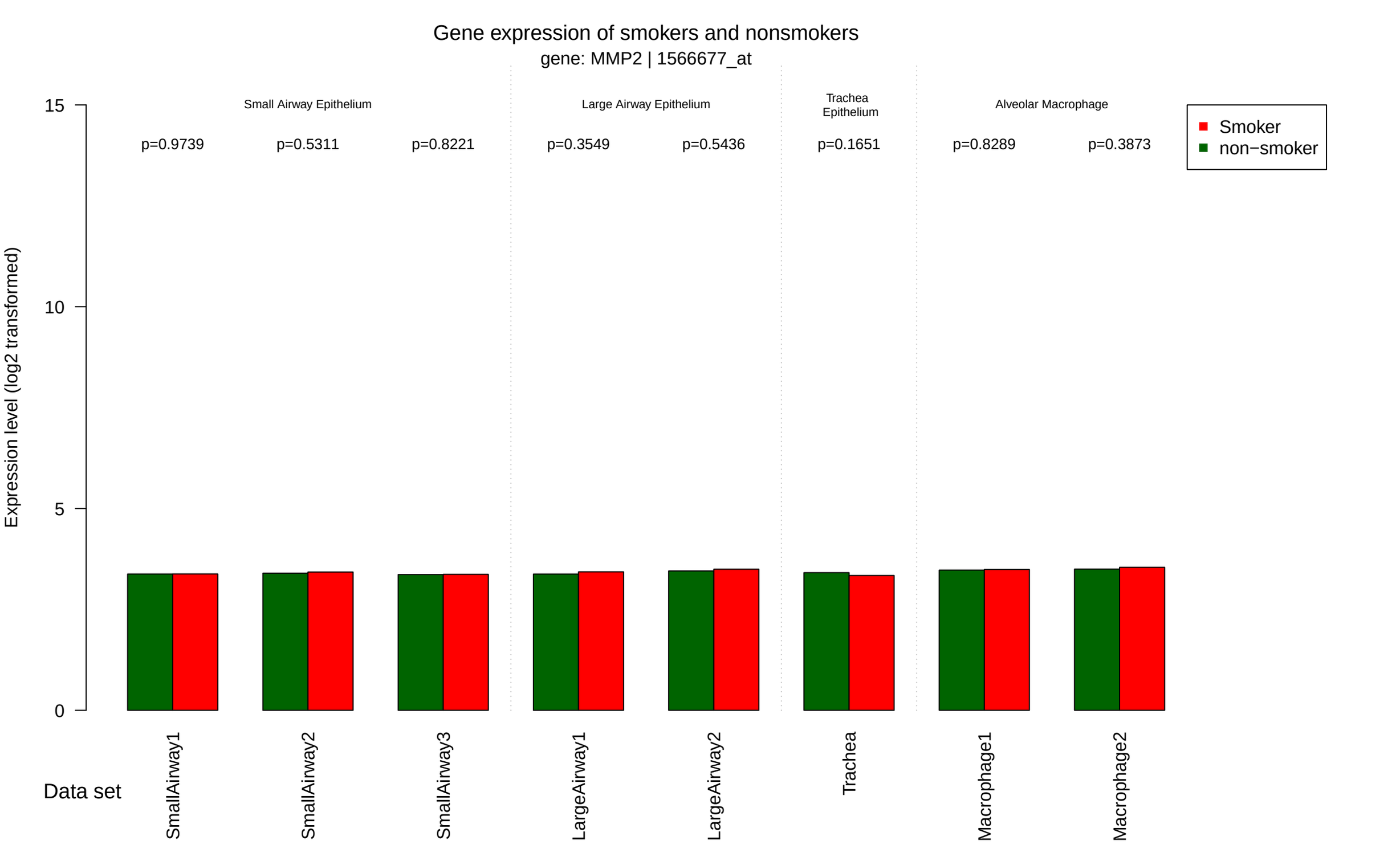
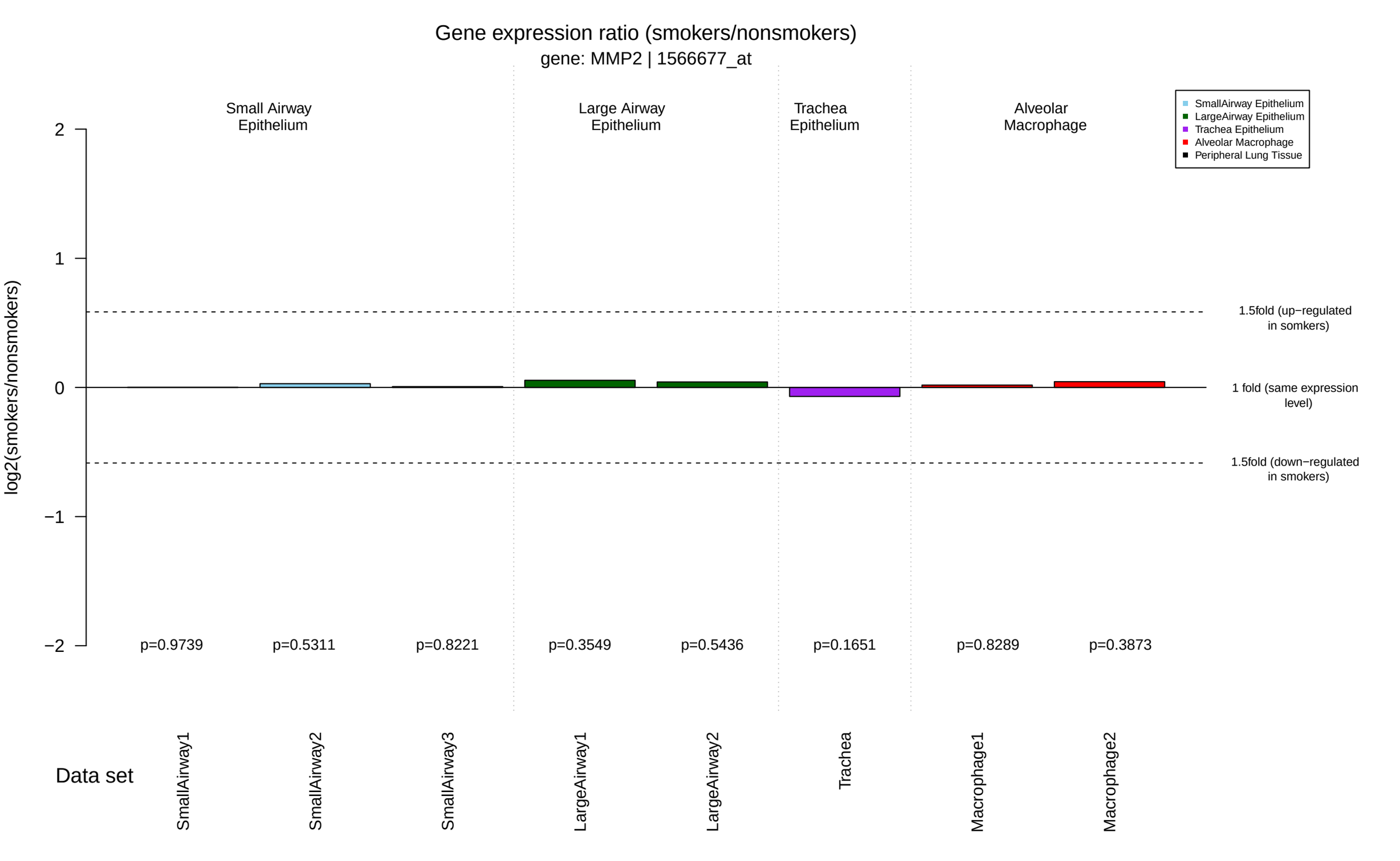
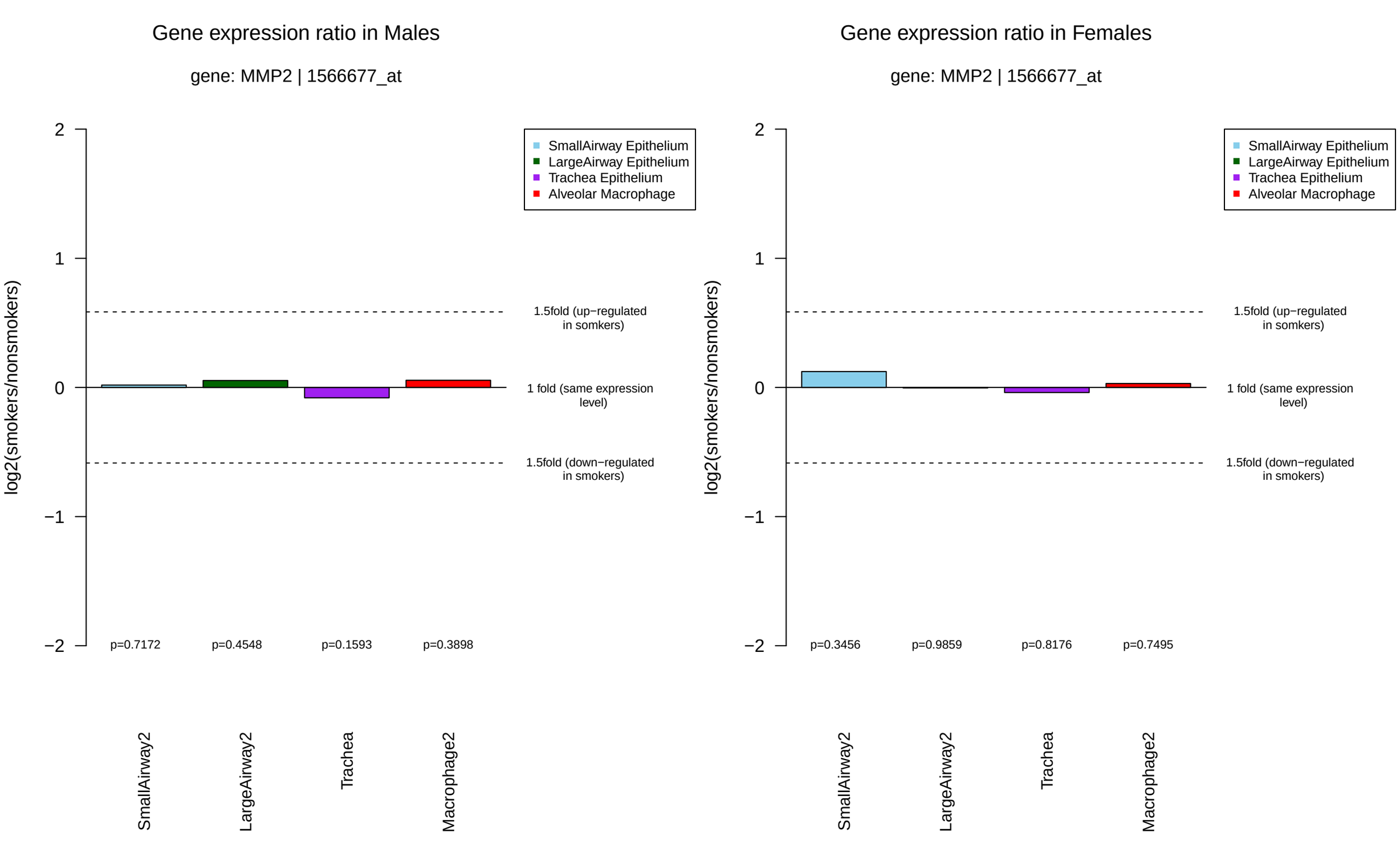
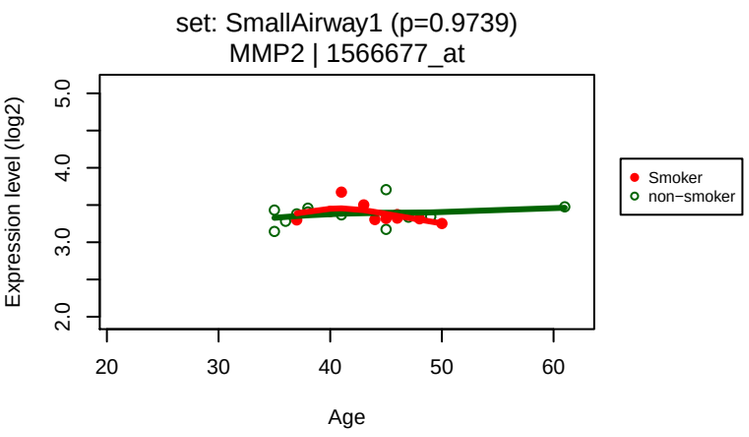
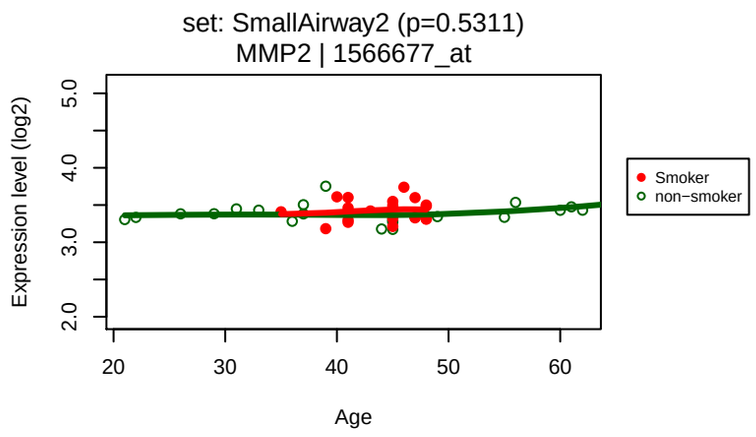
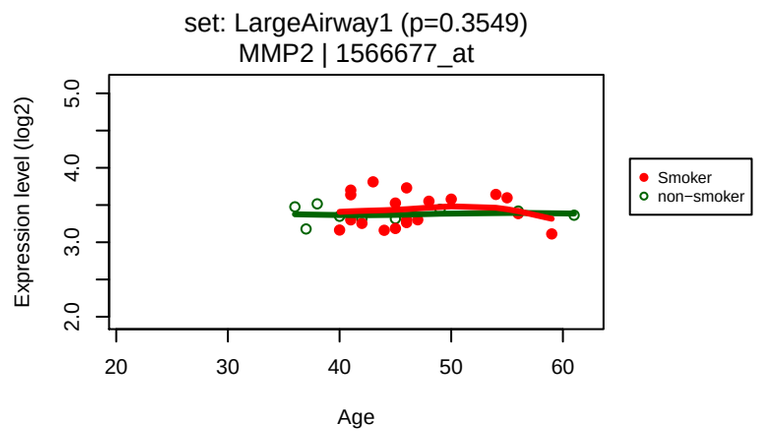

Gene expression of smokers and nonsmokers
gene: MMP2 | 1566677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9739
p=0.5311
p=0.8221
p=0.3549
p=0.5436
p=0.1651
p=0.8289
p=0.3873
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

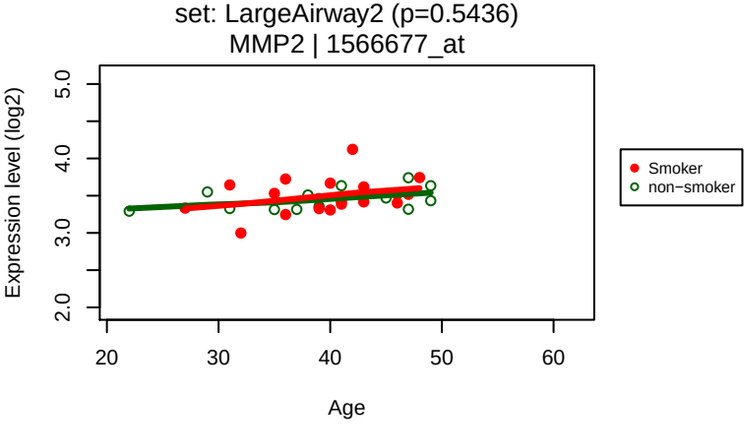

Gene expression ratio (smokers/nonsmokers)
gene: MMP2 | 1566677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9739
p=0.5311
p=0.8221
p=0.3549
p=0.5436
p=0.1651
p=0.8289
p=0.3873
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

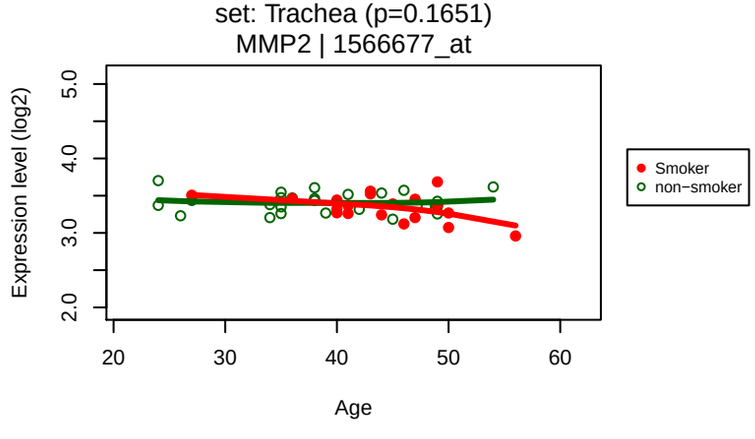

Gene expression ratio in Males
gene: MMP2 | 1566677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7172
p=0.4548
p=0.1593
p=0.3898
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP2 | 1566677_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3456
p=0.9859
p=0.8176
p=0.7495
SmallAirway2
LargeAirway2
Trachea
Macrophage2

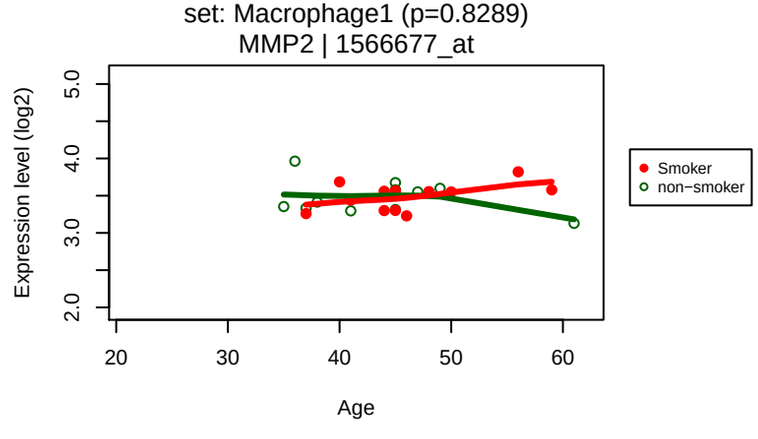

set: SmallAirway1 (p=0.9739)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker

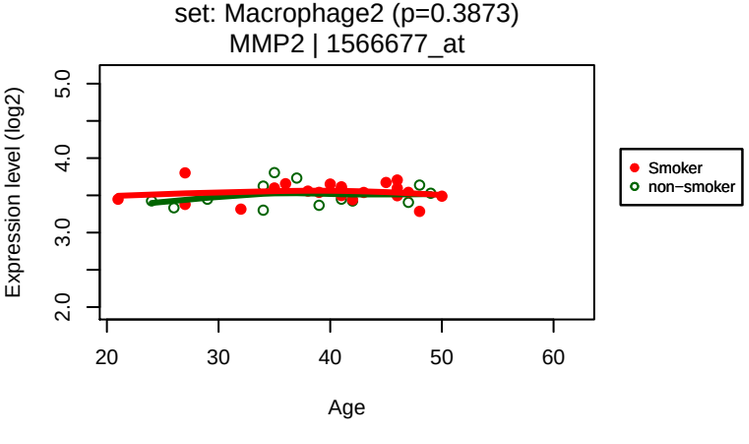

set: SmallAirway2 (p=0.5311)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker

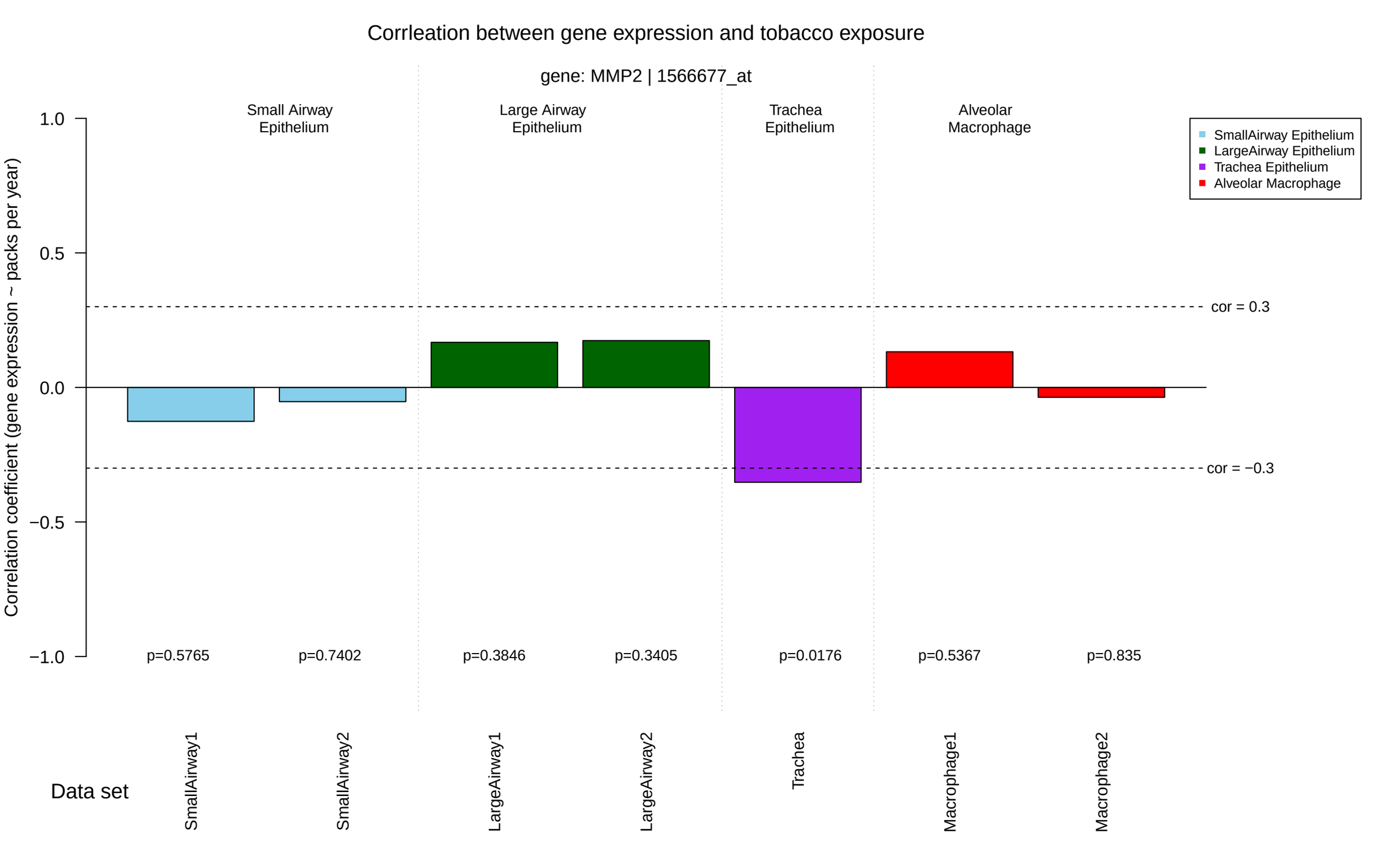

set: LargeAirway1 (p=0.3549)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5436)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1651)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8289)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3873)
MMP2 | 1566677_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP2 | 1566677_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5765
p=0.7402
p=0.3846
p=0.3405
p=0.0176
p=0.5367
p=0.835
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2